Efficient Wake-Word Detection for Low-Resource Languages through Speech Recognizer Distillation

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Abstract

Small always-on wake-word detectors usually need many recordings of each new phrase, which are scarce for low-resource languages. Large speech recognizers model a language well but are too large to run continuously; we aim to transfer their knowledge to a lightweight streaming detector. We propose a two-phase framework: Phase 1 distills a 0.6B-parameter speech recognizer into a 5M-parameter streaming Conformer on multi-domain speech; Phase 2 fine-tunes it on the wake phrase with sub-word connectionist temporal classification targets derived from transcripts, so that partial matches become hard negatives. We score test phrases inside continuous audio. On a Vietnamese wake phrase, with about an hour of phrase-specific recordings, the detector misses 20.5 % of wake phrases at 0.1 false accepts per hour, whereas lightweight keyword-spotting baselines miss 57–85 % and the unadapted teacher over 90 %. Recognizer pretraining, rather than distillation itself, gives most of the gain. The detector runs 30 times faster than real time on one desktop CPU thread; isolated-clip scoring underestimates its false rejections almost threefold. Knowledge from a large recognizer thus makes accurate, efficient wake-word detection feasible with little phrase data; streaming detectors should be evaluated in continuous audio.

CCS Concepts

• **Computing methodologies** → **Speech recognition**; *Neural networks*.

Keywords

connectionist temporal classification, evaluation protocol, keyword spotting, knowledge distillation, streaming Conformer, wake-word detection

1 Introduction

Voice assistants such as Apple's Siri [18, 19] listen for their wake phrase in two stages: a small streaming *first-stage detector*, tuned for high recall, runs all the time on a low-power processor, and a larger *second-stage detector*, tuned for high precision, runs only when the first fires and rejects its false triggers. Both stages are usually trained for every new wake phrase. Small keyword-spotting (KWS) networks [7, 16, 27] are typically trained from scratch on recordings of the phrase, which are expensive to collect, while large automatic speech recognition (ASR) models already encode the sounds of a language but are too large to run continuously. Needing fewer phrase recordings would make new wake phrases, and languages with little wake-word data, cheaper to support. We call this setting low-resource: phrase recordings are few, but general speech data and a pretrained recognizer exist.

We propose a framework that transfers the knowledge of a large ASR model into one lightweight streaming detector (Figure 1). The same detector can run as a high-recall first stage (low threshold) or as a standalone high-precision detector (high threshold). Our contributions are:

- A two-phase training framework: Phase 1 (ASR distillation) compresses an ASR teacher into a small streaming student on multi-domain speech; Phase 2 (wake-word training) adapts it to a wake phrase with limited phrase-specific data.

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ICIIT 2027, Ho Chi Minh City, Vietnam
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ACM ISBN 978-1-4503-XXXX-X/2027/03
https://doi.org/XXXXXXX.XXXXXXX

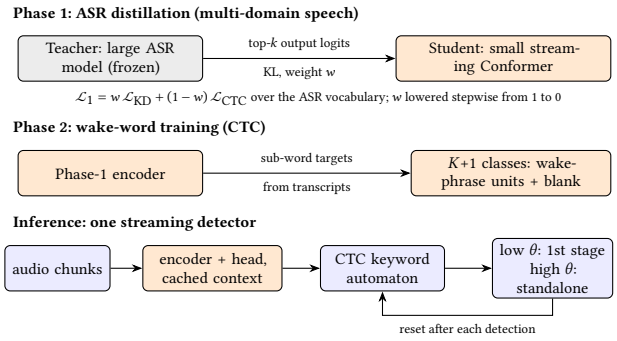


Figure 1: The two-phase framework. Phase 1 distills an ASR teacher into a small streaming student; Phase 2 replaces its output layer with the sub-word units of the wake phrase plus blank; at inference, the threshold θ sets the operating point.

- A rule-based procedure for building sub-word connectionist temporal classification (CTC) [4] targets for a new wake phrase, in which training sentences that partially match the phrase become hard negatives.
- An evaluation protocol that scores each test phrase inside continuous audio rather than as an isolated clip, with near-miss phrases and a sealed long-audio false-accept test.

On one Vietnamese phrase, pretraining the encoder on large multi-domain ASR data gives the largest gain and distillation itself less than the spread between runs; the 5M-parameter detector outperforms untuned depthwise-separable CNN (DS-CNN) and BC-ResNet [7] detectors, openWakeWord [17], the unadapted 0.6B-parameter teacher and teacher-verified cascades at 0.1 false accepts per hour (test-set thresholds); and isolated-clip scoring can greatly underestimate the false rejections of streaming detectors.

2 Related Work

Small-footprint KWS. CNNs [16], DS-CNNs [27] and broadcasted residual networks (BC-ResNet) [7] are typically compared on Speech Commands [25], whose benchmark scores isolated one-second clips. **CTC detectors.** CTC [4] lets a detector emit sub-word units without alignments; streaming Transformers have also been used for wake words [24]. **Transfer and distillation.** Knowledge distillation (KD) [6] has been used to train KWS models from self-supervised speech models [26]; pretrained speech embeddings [10], as in openWakeWord [17], and small ASR models [13] also reduce phrase-specific data. **Vietnamese.** Keyword search in recorded Vietnamese speech has been studied [22]; to our knowledge, always-on wake-word detection has not.

3 Two-Phase Framework

3.1 Phase 1: ASR Distillation

Phase 1 first trains the student as a general streaming recognizer, so that its encoder learns the sounds of the language from large and varied speech corpora rather than from one phrase. The teacher can be any accurate ASR model whose output units the student can share. The student is trained on transcribed utterances with

$$\mathcal{L}_1 = w\,\mathcal{L}_{\mathrm{KD}} + (1-w)\,\mathcal{L}_{\mathrm{CTC}}, \tag{1}$$

where $\mathcal{L}_{\mathrm{KD}}$ is the frame-level Kullback–Leibler divergence between the teacher and student output distributions at temperature τ [6], computed on the teacher's k highest-scoring units per frame; $\mathcal{L}_{\mathrm{CTC}}$ is the CTC loss [4] against the transcript; and w sets their balance. Training starts from pure distillation, which gives dense frame-level targets, and w is lowered stepwise to pure CTC, which ties the student to the transcripts; each new value of w starts when the validation word error rate (WER) stops improving and is reached by a linear transition. Because a general corpus contains few sentences with the wake-phrase words, a fixed share of every batch can be drawn from training sentences that contain them; this optional step, used in our main model, is the only phrase-specific part of Phase 1.

3.2 Phase 2: Wake-Word Training

The distilled encoder is kept and its output layer is replaced by one with K+1 classes: the K sub-word units of the wake phrase and the CTC blank. Training uses CTC on positives (recordings of the phrase) and negatives (other speech, near-miss phrases and noise) in balanced batches. Phase 2 (1) splits the wake phrase into the teacher's sub-word units $u_1 \dots u_K$; (2) writes, for each unit, a rule matching the words likely to be confused with it (same onset or rhyme; short frequent syllables only as whole words); (3) converts every training transcript word by word, a word emitting the units whose rules it matches, so a positive yields $u_1 \dots u_K$, a partially matching negative a partial sequence (a hard negative) and other negatives an empty target; (4) records the phrase and near-miss phrases (several speakers) and perturbs them (speed, pitch, volume, noise, overlapping speech); and (5) fine-tunes the Phase-1 model with CTC on these targets, selecting checkpoints and thresholds on held-out speakers. For example, with the Vietnamese tokenizer used in this work, *năng khiếu ơi* gives four units _năng, _kh, iếu, _ơi (five classes): *năng* and *ơi* are matched only as exact words, a word starting with *kh* emits _kh and a word ending in *iếu* emits iếu, so *khiếu nại* and *nàng khiếu ơi* yield partial targets.

3.3 Streaming Detection

Frame posteriors drive a CTC keyword automaton: a forward recursion that sums all CTC paths through an automaton tracking how much of $u_1 \dots u_K$ the collapsed output has matched, so the phrase may start at any frame; its final (matched) state is absorbing, and the state is renormalized every frame. After each chunk it outputs the log-odds s between the probability mass in the matched state and the rest; a detection fires when $s \geq \theta$, after which the automaton is reset and a 1 s refractory period applies.

4 Experimental Setup

We call our detector the Distilled Conformer and evaluate it on the Vietnamese phrase *năng khiếu ơi*, with the teacher parakeet-ctc-0.6b-Vietnamese [12], a 0.6B-parameter FastConformer-CTC [15] recognizer with 1,024 SentencePiece units.

Student and Phase 1. The student is a cache-based streaming Conformer [5, 11] with 64 log-mel inputs, 8× causal subsampling to 80 ms frames (the teacher's frame rate), five layers of width

192 with 8 attention heads and causal convolutions, and chunked attention over 0.64 s chunks with 5.12 s of cached left context (5,045,829 parameters with the Phase-2 head). Phase 1 uses 1.47M utterances (1,776 h) from nine Vietnamese corpora—InfoRE2, VSV-1100, VLSP, LSVSC [21], FOSD, VietMed [9], Common Voice [1], FLEURS [2] and VAIS-1000—plus MS-SNSD noise [14]; $k = 64$, $\tau = 2$, $w \in \{1, 0.75, 0.5, 0.25, 0\}$ with 2k-step transitions, and 5 % of each batch drawn from sentences containing *năng*, *khiếu* or *ơi*. Our encoder is trained for 262k steps with AdamW (learning rate $5 \cdot 10^{-4}$) on one RTX 4090.

Wake-phrase data and Phase 2. The wake-phrase corpus has 52 speakers (10 female; 44 with a Southern, 5 with a Northern and 3 with a Central accent). Before training the reported models, a fixed random draw stratified by gender and accent split them into 37 speakers for training (7 female), 5 for model selection and 10 for testing. Training uses 1,763 recordings of the phrase (about 48 per speaker, 0.86 h in total) and 514 near-miss recordings (0.24 h), expanded by perturbation to 52k positive clips. Negatives come from the near-miss recordings, InfoRE2, VLSP, LSVSC, ViMD [3] and MUSAN [20] in shares 0.25/0.30/0.15/0.12/0.10/0.08; 36 % of clips pass through measured room impulse responses [8] or telephone filters. Phase 2 trains 10k steps (AdamW, learning rate 10^{-4}).

Test protocol. The 10 test speakers (2 female; 2 with a Northern and 1 with a Central accent) give 417 recordings of the phrase and 119 near-miss recordings (*nàng khiếu ơi*, *năng khiếu âm nhạc*, …; 57 of the 65 near-miss texts are also spoken by training speakers), each in up to 20 versions (the original, missing for one speaker's 79 recordings, plus speed, pitch, volume, noise and overlap variants): 7,578 positive and 2,345 near-miss clips. Each clip is placed after 20.48 s of an 11.76 h Vietnamese livestream, which continues under the clip 20 dB below it, so the detector state is warm as in continuous use ("in context"). False accepts are counted by streaming 111.5 h of audio continuously: the livestream and a sealed set of 105 YouTube videos from 75 channels (nine genres, e.g., game shows, sports, music and drama; 99.75 h). Because VSV-1100 is web-crawled, its 1,020,902 utterances were matched against the false-accept audio by landmark fingerprinting [23]: they overlap 0.08 % of the livestream and 0.44 % of the sealed set, where overlapping regions are not counted. The deployment threshold (see model selection) is applied once to the sealed set before any other measurement of these models on it.

Metrics. Let s_i be a clip's highest detection score (for our detector, the log-odds s) inside its window (the clip plus one chunk). For a threshold θ, the false rejection rate $\mathrm{FRR}(\theta)$ is the fraction of positives with $s_i < \theta$, the near-miss accept rate $\mathrm{NM}(\theta)$ the fraction of near misses with $s_i \geq \theta$, recall = 1 − FRR, and $\mathrm{FA/h}(\theta) = E(\theta)/T$ for $E(\theta)$ detections on T hours of continuous audio. "FRR at x FA/h" is the lowest $\mathrm{FRR}(\theta)$ with $\mathrm{FA/h}(\theta) \leq x$ and $\mathrm{NM}(\theta) \leq 0.1$, so that recall cannot be bought by accepting near misses. FRR intervals come from a bootstrap that resamples speakers and, within them, source recordings; FA/h intervals are exact Poisson. On 111.5 h, 0.1 FA/h allows at most 11 false accepts.

Model selection. Checkpoints are selected by FRR at 1 FA/h and NM≤0.1 on the selection speakers and 4.87 h of other YouTube shows; the threshold of this operating point is the deployment threshold. The Phase-1 encoder, Phase-2 recipe and learning rate were fixed in earlier experiments on another split, in which one current test speaker served for checkpoint selection; nothing was tuned afterwards. A variant that also trains with context insertion (other audio placed before half of the positive and near-miss clips, plus noise and room responses on 30 % of them) is reported as "+ context insertion"; the selection rule prefers the recipe without it.

Table 1: Test FRR at fixed false-accept rates on 111.5 h (10 test speakers, near-miss accepts ≤10 %, thresholds set on the test audio). NM: near-miss accept rate at 0.1 FA/h. Distilled Conformer: mean and range of three Phase-2 runs; other rows: one run.

System	Params	FRR at FA/h				NM
		0.1	0.5	1	2	
Distilled Conformer	5.0M	**0.205**	**0.195**	0.195	0.195	0.093
range		.19–.21	.18–.21	.18–.21	.18–.21	
+ context insertion	5.0M	0.995	0.243	**0.133**	**0.078**	0.000
DS-CNN[a]	137k	0.770	0.272	0.272	0.272	0.011
DS-CNN[a]	1.2M	0.846	0.718	0.642	0.622	0.029
BC-ResNet-8	0.32M	0.592	0.373	0.292	0.223	0.017
openWakeWord[c]	0.51M	0.574	0.340	0.285	0.223	0.020
Teacher, folded posteriors	600M	0.935	0.906	0.894	0.874	0.001
Cascade, folded verifier	605M	0.599	0.599	0.599	0.599	0.099
Text match[b] (teacher or cascade): FRR 0.938 at 0 FA/h, NM 0.000						

[a]With or without context insertion, whichever has the lower FRR at 0.1 FA/h (137k: with; 1.2M: without). [b]Greedy decoding, exact phrase match; stage 1 passes all clips the teacher matches. [c]Frozen 0.33M embedding [10]; DNN or LSTM head with the lower FRR at 0.1 FA/h (LSTM).

Baselines. Causal DS-CNNs in the style of [27] (137k and 1.2M parameters: strided convolution stem to 80 ms frames, six dilated depthwise-separable blocks, the same five-class CTC head, automaton and selection rule, no ASR pretraining) are trained on the same positives and negatives, sampled uniformly instead of with the shares above, with and without context insertion (AdamW, learning rate $2 \cdot 10^{-3}$, 20k steps; not tuned). openWakeWord [17] classifies 16 frames (about 2 s of audio) of a frozen speech embedding [10] with a small DNN or LSTM trained and selected by its own recipe on the same clips, windows ending where the phrase ends (found by forced alignment with the teacher where speech follows the phrase). BC-ResNet-8 [7] classifies 1.5 s windows of 40 log-mel bands every 80 ms, trained from scratch with its paper's optimizer and augmentation on windows built the same way, and selected by our rule. The teacher runs offline without adaptation, either with its posteriors folded to the five classes (the four wake units keep their probability, all else becomes blank) or by greedy decoding and exact phrase matching. Two-stage cascades run our detector at 10 FA/h and let the teacher verify, with either scoring, the audio from 3 s before to 0.64 s after each detection (for test clips, around the whole clip); text matching alone decodes 4 s windows every 2 s of the false-accept audio.

5 Results and Discussion

5.1 Main Results

With thresholds set on the test audio (Table 1), the Distilled Conformer misses 20.5 % of in-context wake phrases at 0.1 FA/h (runs

Table 2: Phase-1 encoders and partial targets: test FRR on 111.5 h (near-miss accepts ≤10 %); Phase 2 is fixed unless noted. NM at 0.1 FA/h. *Mean of three Phase-2 runs on one Phase-1 encoder.

Phase-1 encoder	FRR at FA/h				NM
	0.1	0.5	1	2	
None (random initialization)[d]	0.914	0.879	0.879	0.879	0.066
ASR, single domain (InfoRE2, 416 h)	0.523	0.402	0.402	0.402	0.055
ASR, 9 corpora, CTC only*	0.258	0.227	0.227	0.227	0.079
ASR, 9 corpora, distillation*	0.227	0.199	0.199	0.199	0.081
+ transitions, wake-word sentences*	**0.205**	**0.195**	**0.195**	**0.195**	0.093
Same encoder, whole-phrase targets*	0.268	0.222	0.222	0.222	0.077

[d]Encoder learning rate 10^{-3}.

19.4, 21.3 and 20.9 %; 95 % CI of the first run 11.8–26.5 %; mean 20.6 % without the test speaker used in development) and hardly fewer at 2 FA/h: the near-miss cap, not the false-accept budget, sets the threshold. For the first run (designated in advance; best of three), the most accepted near misses share most of the phrase (*nàng khiếu ởi* 75 %, *năng khiếu âm nhạc* 50 %), and FRR is 0.089 on the 338 unperturbed clips and 0.05–0.34 per speaker (0.23 and 0.20 for the two female speakers). The deployment threshold, fixed on selection data, gives a test FRR of 0.096 at 1.79 FA/h on the livestream but accepts 25 % of near misses, possibly because the selection speakers have only 52 near-miss recordings; on the sealed set it gives **0.38 FA/h** [0.27, 0.52] (38 false accepts in 99.75 h; 0–0.96 FA/h per genre except 1.9 on 1.1 h of drama). The context-insertion variant has the lowest FRR at 1–2 FA/h but misses 99.5 % at 0.1 FA/h, and gives 1.00 FA/h on the sealed set. All baselines miss more at every operating point of Table 1; BC-ResNet, which accepts fewer near misses, falls below our near-miss-limited FRR only above 2.9 FA/h (Figure 2). Even on unperturbed clips decoded alone, the unadapted teacher misses 78 %, often mis-transcribing the short phrase (e.g., *nhặng thiếu rồi*).

5.2 Effect of the Phase-1 Encoder

Without ASR pretraining (Table 2), the FRR at 0.1 FA/h is 0.914. One read-speech corpus (416 h, distillation) lowers it to 0.523; nine corpora (1,776 h) with CTC only to 0.258 (three runs: 0.221–0.284); distillation with abrupt steps of w, still without phrase-specific data, to 0.227 (0.196–0.265); and linear transitions between the steps plus wake-word sentences (two changes, not separated) to 0.205 (0.194–0.213), the lowest mean and the smallest spread between runs. ASR pretraining on more, and more varied, data is thus the largest factor; abrupt-step distillation (252k vs 116k Phase-1 steps; the CTC-only WER had plateaued) lowers the mean FRR of CTC alone at every operating point, although the run ranges overlap. Partial targets also matter: emitting units only for complete phrases raises FRR to 0.268 (0.257–0.282).

5.3 One Model, Two Operating Points

As a first stage, the detector (first run) keeps 95.1 %, 97.2 % and 99.2 % of wake phrases at 10, 30 and 100 FA/h on the livestream (no near-miss cap), waking a verifier for 1.0, 3.0 and 10.0 % of the

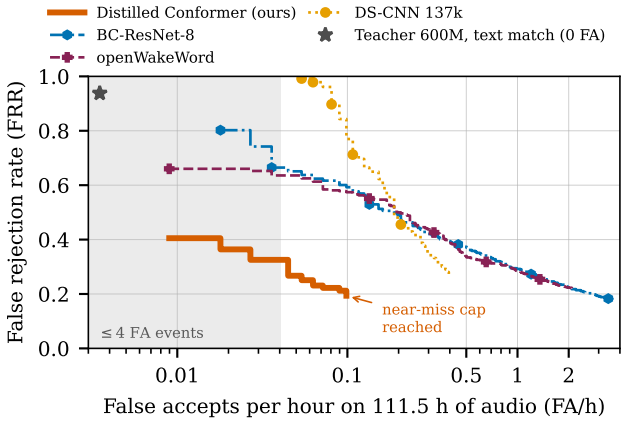


Figure 2: Detection error trade-off (DET) curves on 111.5 h of multi-domain audio for our detector (first run) and the strongest baselines of Table 1 (near-miss accepts ≤10 %; curves end where this cap binds). Gray band: ≤4 false accepts; 0-FA point at the left edge.

Table 3: Test clips scored in isolation and in context at ≤1 FA/h on the livestream alone (≤11 false accepts, as at 0.1 FA/h in Table 1; no near-miss cap; coarser grid). Cold: share of 2,000 random 5.12 s livestream segments triggering a reset detector.

Detector	FRR		NM		Cold
	isolated	context	isolated	context	
Distilled Conformer	0.047	0.129	0.153	0.185	0.022
Single-domain encoder	0.047	0.307	0.215	0.151	0.048
earlier Phase-2 recipe[e]	0.003	0.165	0.980	0.839	0.830
DS-CNN 137k	0.045	0.864	0.786	0.032	0.026
with context insertion	0.791	0.779	0.004	0.011	0.000

[e]Encoder frozen for the first 2k steps, then learning rate 10^{-5}.

audio. Adding the unadapted 0.6B teacher as a second stage did not improve on the single model: at 0.1 FA/h the cascade with folded posteriors misses 59.9 % (19.4 % for the same model alone); with text matching it has no false accepts but inherits the teacher's FRR. A phrase-trained verifier was not tested.

5.4 Isolated Clips versus In-Context Scoring

Table 3 scores the same clips in isolation and in context; in isolation each clip, followed by 0.64 s of silence, is scored from a reset state. For our detector (first run) and the current-recipe single-domain model and DS-CNN, all without context insertion, isolated scoring underestimates the in-context FRR 2.7 to 19 times; with the background 30 dB down, our in-context FRR is still 0.110 (Section 5.5), so the gap is not mainly due to the background. Isolated scoring can also hide large differences: the DS-CNN matches our detector on isolated clips (4.5 vs 4.7 %) but misses 86 % in context, and a single-domain model from an earlier Phase-2 recipe misses 0.3 % of isolated clips but fires on speech onset after a reset (on 83 % of random segments). Training with context insertion removes the gap

for the DS-CNN, but at 78–79 % FRR. Streaming detectors should therefore also be evaluated in continuous audio.

5.5 Robustness and Efficiency

For the first run, FRR at 1 FA/h on the livestream (new insertion positions, no near-miss cap) is 0.110, 0.125 and 0.149 with the background 30, 20 and 10 dB below the speech, and 0.217 with held-out MS-SNSD test noise at 10 dB SNR. On isolated clips of the selection and test speakers (reset state; false accepts counted as triggered 10 s livestream segments, so only relative changes matter), FRR at 1 FA/h rises from 0.168 without channel effects to 0.373 over a telephone channel and 0.352–0.594 with reverberation times of 0.2–0.8 s: telephone and far-field use need further work. Chunking adds up to 0.64 s of latency. The model needs 19.2 MiB in fp32 and 4.8 MiB with per-channel int8 weights (simulated: FRR 0.195 vs 0.194 in fp32 at 1 FA/h on the livestream) plus 0.26 MiB of encoder cache, and 0.22 GFLOPs per second of audio with cached context; recomputing the 5.12 s context for every 0.64 s chunk takes a median of 21 ms on one thread of a desktop Core i7-13700K (PyTorch 2.6), 30 times faster than real time. Mobile hardware was not measured.

6 Conclusion

We distilled a large ASR model into a streaming wake-word detector trained with sub-word CTC targets and scored it in continuous audio. On one Vietnamese phrase, with 52 minutes of phrase recordings, the 5M-parameter detector misses 20.5 % of wake phrases from ten test speakers at 0.1 FA/h (test-set thresholds), beats untuned convolutional detectors, openWakeWord, the unadapted teacher and teacher-verified cascades up to 2 FA/h, and runs 30 times faster than real time on one desktop CPU thread; at a threshold fixed on held-out data it gives 0.38 FA/h on 100 h of unseen audio. ASR pretraining gives most of this accuracy and partial targets improve it; distillation lowers the mean FRR by less than the spread between runs; isolated-clip scoring understates the errors of streaming detectors. The evaluation covers one phrase, ten test speakers and simulated audio. Near misses that share most of the phrase remain the main difficulty (25 % accepted at the held-out threshold); other phrases, languages with little transcribed speech, far-field use and mobile hardware are future work.

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